# BT 201 Mid Current Solved

**Systematics?**
systematics (cladistics) is a method of taxonomic classification based on their evolutionary history. It was developed by Willi Hennig, a German entomologist, in 1950.

**Species richness Systematics?**
Species richness is the number of different species in a particular community. If we found 30 species in one community and 300 species in another, the second community would have much higher species richness than the first.

**Phylogeny?**
Phylogeny is the evolutionary history of a species or group of related species  The discipline of systematics classifies organisms and determines their evolutionary relationships.

**Difference between the Arctic and polar region?**
the **Arctic is** not a continent which **is** the predominant difference **between** the two **polar regions**. Under the massive ice cap **of** the **Arctic** lies the **Arctic Ocean** whose depths reach 15,000 feet below the surface. Often times the two **polar regions** are confused due to their having such similar names. **(Internet)**

**Law of succession?**
General correspondence between fossil and living forms in the same geographical area • Darwin used this observation to predict that fossils of ancient humans should be found in Africa

**Bioinformatics systematics?**
**???**

**Why breeding crops important?**
**Crop breeding** is the art and science of improving **important** agricultural plants for the benefit of humankind. **Crop breeders** work to make our food, fiber, forage, and industrial **crops** more productive and nutritious. **Crops** provide for an expanding global population with increasing dietary expectations. **(Internet)**

**Does factor affect commute structure?**
The gender, age, number of family cars, housing source, and jobs-housing balance are the most important **factors** influencing **commuting** mode choice; housing, population density, jobs-housing balance, and **commuting** mode significantly **affect commuting** time; and transport accessibility, jobs-housing balance, and **commuting** mode **(Internet)**

**Ecology?**
Ecology is a branch of biology that studies the interactions among organisms and their biophysical environment, which includes both biotic and abiotic components.

"Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).

**Applied Ecology?**
Applied Ecology: Using ecological principles to maintain conditions necessary for the continuation of present day life on earth.

**Plants breeding?**
Plant breeding is the art and science of changing the traits of plants in order to produce desired characteristics. Plant breeding can be accomplished through many different techniques ranging from simply selecting plants with desirable characteristics for propagation, to methods that make use of knowledge of genetics and chromosomes, to more complex molecular techniques.

**Protozoa life cycle?**

**Protozoa** have two-phase **life cycles**, alternating between proliferative stages (e.g., trophozoites) and dormant cysts. As cysts, **protozoa** can survive harsh conditions, such as exposure to extreme temperatures or harmful chemicals, or long periods without access to nutrients, water, or oxygen for periods of time. **(Internet)**

**Five scientist citing reteach?**

- **Kerner** showed that the differences in growth form observed by him and Bonnier were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants.
- **Turesson's** studies provided evidence for genetic differences among populations.
- **Clausen, Keck, and Hiesey** explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.
- **Case** determined that the best predictor of chuckwalla, Saur malus, body length was average winter rainfall  **Tracy's** laboratory growth experiments indicated that variation in body size among chuckwalla populations is at least partly determined by genetic differences among populations.

**How does gene find sustainability of gene?**

**???**

**How climate affect community?**

**Climate** change could **affect** our **society** through impacts on a number of different social, cultural, and natural resources. For example, **climate** change could **affect** human health, infrastructure, and transportation systems, as well as energy, food, and water supplies. **(Internet)**

**Sequencing?**

DNA **sequencing** is the process of determining the nucleic acid sequence – the order of nucleotides in DNA. It includes any method or technology that is used to determine the order of the four bases: adenine, guanine, cytosine, and thymine.

## Long 5 Marks

**Why we use phylogenic tree? Why phylogenic tree is important?**

**Phylogenetics** is important because it enriches our understanding of how genes, genomes, species (and molecular sequences more generally) evolve.

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**Biodiversity in food and health?**

**Biodiversity** creates resilience and is a key to mitigate the risks in agriculture particularly farming, from where we get most of our **food**. It ensures the world gets nutrients which is critical for good **health** and is also critical for conserving the ecosystem that produces the **food**. **(Internet)**

**Factor forming community shape?**

**Shape The Forces That Bring A Community Together**

- Common goals. Common goals (whether long-term or short-term) unite members to contribute towards it. ...
- Common threat. Common threats, problems and enemies help forge groups together. ...
- Isolation from others. Isolation from others focuses the attention upon each other. ...  Barriers to entry. Barriers to entry are important. **(Internet)**

**Landscape ecology explain?**

**Landscape ecology** is the science of studying and improving relationships between **ecological** processes in the environment and particular ecosystems. This is done within a variety of **landscape** scales, development spatial patterns, and organizational levels of research and policy. **(Internet)**

**Linnaeus classification?**

Carolus Linnaeus is the father of taxonomy, which is the system of classifying and naming organisms. One of his contributions was the development of a hierarchical system of classification of nature. Today, this system includes eight taxa: domain, kingdom, phylum, class, order, family, **genus**, and **species** **(Internet)**

**Phylum Annelida?**

**Annelid**, **phylum** name **Annelida**, also called segmented worm, any member of a **phylum** of invertebrate animals that are characterized by the possession of a body cavity (or coelom), movable bristles (or setae), and a body divided into segments by transverse rings, or annulations, from which they take their name. **(Internet)**

Phylum Annelida Class Polychaeta Family Siboglinidae Class Clitellata • Subclass Oligochaeta. Subclass Hirudinea

**Annelid Characteristics** • Defining Characteristics • One or more pairs of chitinous setae • The phylum includes polychaetes, earthworms, leeches, and vestimentiferans • True segmented worms • Metameric segmentation

**Body Structure** The body is a tube within a tube. The coelom is important to annelids for:The epidermis is what secretes the tough cuticle

**Locomotion** On each side of the animal is a parapod (parapodia) consisting of fleshly lobes, which are supported by chitinous rods. Each parapod have setae, which can be sharp (protection), and aid in locomotion

**Feeding** Annelids range from carnivores, herbivores, scavengers, deposit feeders, and filter feeders. With very few defenses, many remain in a burrow or secreted tube. Carnivores can capture prey with strong jaws and quickly drag it back to its burrow. Can use a muscular pharynx = eversible proboscis. **(PPTS)**